

Network Troubleshooting, Management, Monitoring, and Optimization

Task 1: Use the Windows command prompt to run the command `ipconfig /all` on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.

```
Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::9b69:b498:ce21:5f25%12
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2405:201:200f:c1cf:8809:a17c:6516:6cf4
    Temporary IPv6 Address. . . . . : 2405:201:200f:c1cf:7869:645b:f8a4:8f0d
    Link-local IPv6 Address . . . . . : fe80::7bb7:f9b4:f160:a70e%3
    IPv4 Address. . . . . : 192.168.29.218
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::4e93:a6ff:fe8d:56c6%3
                                192.168.29.1
```

Question 2: Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

```
C:\Windows\System32>ping www.google.com
Ping request could not find host www.google.com. Please check the name and try again.

C:\Windows\System32>_
```

1. Disconnected State:

- Command Output: Ping request could not find host www.google.com. Please check the name and try again or Request timed out.
- Status Meaning: Indicates physical layer failure / medium disconnect. The computer cannot reach local gateway or DNS to resolve the target server name.

2. Reconnected State:

- Command Output: Reply from 142.250.193.206: bytes=32 time=22ms TTL=116
- Status Meaning: Indicates successful end-to-end network connectivity layer establishment across physical, transport, and application routing pathways.

```
C:\Windows\System32>ping www.google.com

Pinging www.google.com [2001:4860:4827:7700::] with 32 bytes of data:
Reply from 2001:4860:4827:7700::: time=26ms
Reply from 2001:4860:4827:7700::: time=21ms
Reply from 2001:4860:4827:7700::: time=25ms
Reply from 2001:4860:4827:7700::: time=22ms

Ping statistics for 2001:4860:4827:7700:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 21ms, Maximum = 26ms, Average = 23ms

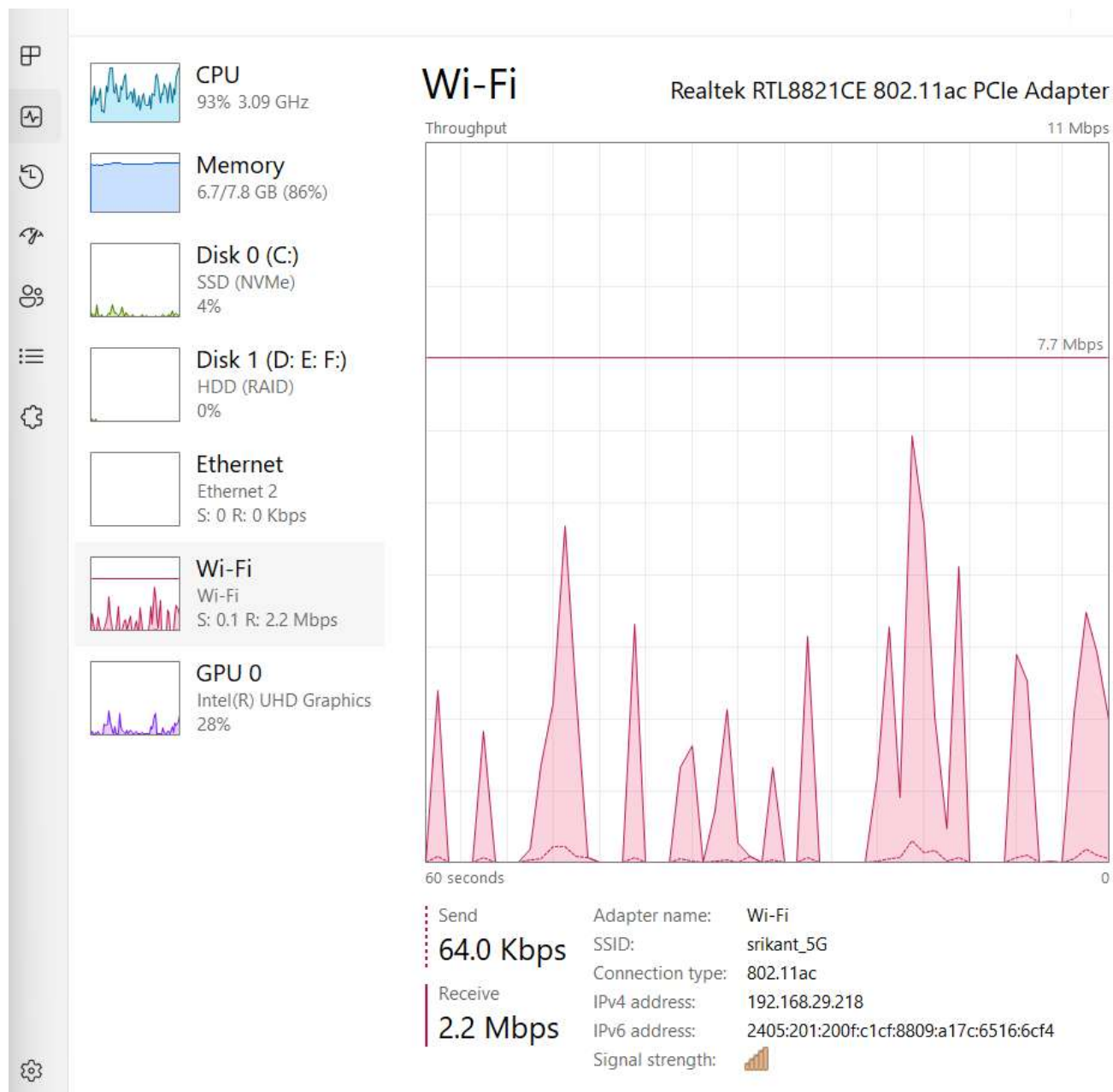
C:\Windows\System32>_
```

Task 3: Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.

Resource Monitor Findings:

During a 5-minute YouTube streaming session, peak network throughput on the Wi-Fi adapter reached approximately 11 Mbps, with receive rates fluctuating between roughly 0.5-11 Mbps depending on

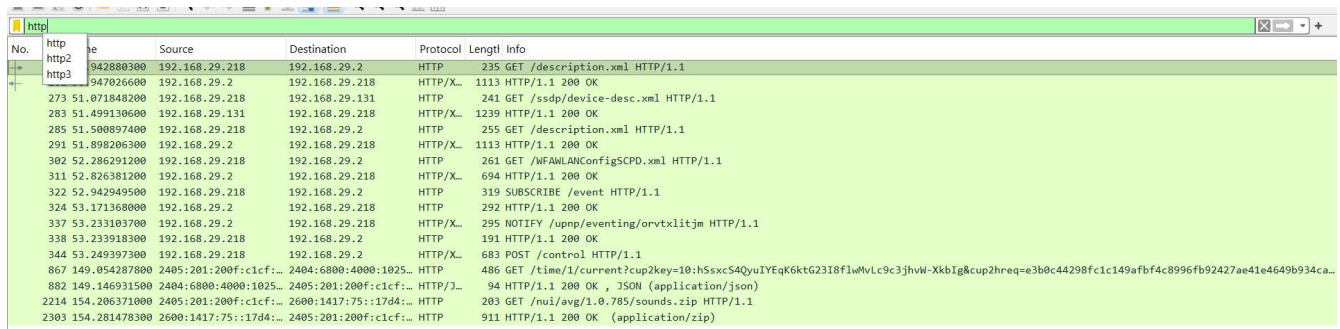
buffering and video quality, while send traffic stayed minimal (under 100 Kbps, mostly acknowledgment/control traffic).



Task 4: Download and install Wireshark (or use an online packet capture demo). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets.

After filtering HTTP/HTTPS traffic in Wireshark, the following three types of information can be observed:

1. Source and destination IP addresses of communicating devices.
2. HTTP request methods and requested resources (e.g., GET, POST, URLs).
3. Server response status codes and content types (e.g., 200 OK, JSON, ZIP).



No.	Time	Source	Destination	Protocol	Length	Info
2	0.000000	192.168.29.218	192.168.29.2	HTTP	235	GET /description.xml HTTP/1.1
3	0.000000	192.168.29.2	192.168.29.218	HTTP/X...	1113	HTTP/1.1 200 OK
273	51.071848200	192.168.29.218	192.168.29.131	HTTP	241	GET /ssdp/device-desc.xml HTTP/1.1
283	51.499130600	192.168.29.131	192.168.29.218	HTTP/X...	1239	HTTP/1.1 200 OK
285	51.500097400	192.168.29.218	192.168.29.2	HTTP	255	GET /description.xml HTTP/1.1
291	51.898206300	192.168.29.2	192.168.29.218	HTTP/X...	1113	HTTP/1.1 200 OK
302	52.286291200	192.168.29.218	192.168.29.2	HTTP	261	GET /wfaWLANConfig5CPD.xml HTTP/1.1
311	52.826381200	192.168.29.2	192.168.29.218	HTTP/X...	694	HTTP/1.1 200 OK
322	52.942949500	192.168.29.218	192.168.29.2	HTTP	319	SUBSCRIBE /event HTTP/1.1
324	53.171368000	192.168.29.2	192.168.29.218	HTTP	292	HTTP/1.1 200 OK
337	53.233103700	192.168.29.2	192.168.29.218	HTTP/X...	295	NOTIFY /upnp/eventing/orvtxlitjm HTTP/1.1
338	53.233918300	192.168.29.218	192.168.29.2	HTTP	191	HTTP/1.1 200 OK
344	53.249397300	192.168.29.218	192.168.29.2	HTTP/X...	683	POST /control HTTP/1.1
867	149.054287800	2405:201:200f:c1cf::2404:6800:4000:1025::	2405:201:200f:c1cf::	HTTP	486	GET /time/1/current?cup2key=10:hSxxc54QyuIYEgK6ktG23I8f1wVlc9c3jhwW-XkbIg&cup2hreq=ae3b0c44298f1c149afb4c8996fb92427ae41e4649b934ca...
882	149.146931500	2404:6800:4000:1025::	2405:201:200f:c1cf::	HTTP/1...	94	HTTP/1.1 200 OK , JSON (application/json)
2214	154.206371000	2405:201:200f:c1cf::	2600:1417:75::17d4::	HTTP	203	GET /nu/avg/1.0.785/sounds.zip HTTP/1.1
2303	154.281478300	2600:1417:75::17d4::	2405:201:200f:c1cf::	HTTP	911	HTTP/1.1 200 OK (application/zip)

ut slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.

Switch from Wi-Fi to a wired Ethernet connection — Wi-Fi introduces latency, jitter, and packet loss from interference and signal drop, all of which hurt gaming performance far more than they hurt regular browsing. A wired connection gives the most stable, lowest-latency link to the router.

1. **Restart the router/modem and check for firmware updates**
2. **Reduce network congestion from other devices** — Streaming, large downloads, or cloud backups running on other devices during a gaming session eat into available bandwidth. Pausing these on the router to prioritize gaming traffic, helps keep latency low.
3. **Connect to the correct Wi-Fi band and reduce interference** — If wired isn't possible, use the 5GHz band instead of 2.4GHz (less interference, lower latency, though shorter range), and keep the router away from microwaves, cordless phones, and thick walls.
4. **Check ping/latency to the game server, not just download speed** — High download speed doesn't guarantee low latency. Use ping or an in-game network diagnostic to check round-trip time; latency above ~50-80ms or inconsistent "jitter" is often the real culprit behind lag, not raw bandwidth.
5. **Contact the ISP or check for local network congestion** — If none of the above help, run a speed test at different times of day; consistent slowdowns during peak hours can indicate ISP-side congestion rather than a home network issue.

